

# Research Execution Blueprint for a Background-Adjusted, Cross-City ATM-Site Crime Study

## Executive summary

The attached material supports a clear redesign: **stop extending the earlier SEM-decision-tree-SHAP-FIS-Choquet pipeline and instead build a rigorously defined, exposure-aware ATM-site prediction study in New York City, Buffalo, and Rochester**. The central question should be whether the immediate environment of an ATM site contains predictive information **beyond ordinary surrounding crime**, and whether that incremental signal survives transfer to another city. The longer 29-page document contains the full rationale and design; the shorter attachment substantially restates its later execution and fallback plan. [filecite:turn0file1](#) [filecite:turn0file0](#)

The strongest contribution is therefore not a novel classifier. It is a clean empirical test of three things: **incremental ATM-site risk, external transportability, and the value of limited target-city data**. This direction is well aligned with prior cross-city crime research: Bappee et al. explicitly treated Toronto and Vancouver as source domains and Halifax as a target domain, found source-only transfer weaker than configurations incorporating target examples, and documented the possibility of negative transfer when domains were dissimilar. <sup>1</sup> More sophisticated unsupervised cross-city adaptation has also been demonstrated, so there is no need to make deep domain adaptation the first version of this paper. <sup>2</sup> Cross-jurisdictional place-based research likewise finds that environmental risk factors may recur across cities without having identical effects, which is precisely the scientific pattern this ATM study should test.

<sup>3</sup>

The **largest unresolved issue is ATM exposure history**. The New York State Department of Financial Services dataset is an excellent common denominator because it is statewide, has individual ATM locations, is posted monthly, and specifically covers bank-owned or in-bank ATMs. However, its public metadata describes the time period as **current**, not as a historical panel, and explicitly excludes independent non-bank ATMs such as machines in delis and bodegas. As of September 15, 2026, that dataset had been updated during the month immediately preceding this report. <sup>4</sup> A retrospective ATM-year study must therefore first determine whether past monthly snapshots can be recovered. A current September 2026 ATM list must **not** simply be assumed to represent the ATM universe in 2022–2025.

The second critical issue is outcome harmonization. The paper should study **ATM-site acquisitive crime risk**, not claim to identify literal attacks on ATM machines unless police records support that interpretation. NYPD records are complaints, with detailed offense descriptors and location fields, and their own documentation notes that complaint classifications reflect the most serious offense and that geocoding/report timing have limitations. <sup>5</sup> Buffalo exposes incident date/type/description fields, while Rochester publicly exposes Part I categories such as robbery, burglary, and larceny. <sup>6</sup> Consequently, a common codebook must be created **before model fitting**, with fraud, forgery, account compromise, and other nonphysical mechanisms separated from the primary endpoint.

The recommended execution has seven gated phases:

| Phase                      | Required output                                          | Main decision gate                                                           |
|----------------------------|----------------------------------------------------------|------------------------------------------------------------------------------|
| Freeze construct           | Signed/preregistered study specification                 | Is one common outcome and unit defensible across cities?                     |
| Build ATM-site panel       | Exposure-aware ATM-period denominator + incident linkage | Can ATM activity periods be reconstructed adequately?                        |
| Build common features      | Identical ex-ante feature schema in every city           | Are enough common variables available without temporal leakage?              |
| Test incremental signal    | M0 vs M1/M2 on strict held-out data                      | Do environmental variables add precise, meaningful predictive value?         |
| Test city transfer         | Zero-shot, multi-source, 10%/25% adaptation              | Is there transportability or measurable transfer benefit?                    |
| Interpret validated models | Stable versus city-specific effects                      | Do explanations replicate in direction/shape?                                |
| Optional placement layer   | Pareto/risk-based decision support                       | Are genuine candidate sites, demand, accessibility, and cost data available? |

The attached plan specifies the **research field, broad objectives, cities, proposed model families, transfer concept, preferred metrics, radii, common-feature philosophy, and fallback hierarchy**. It does **not** fully specify the exact primary crime codebook, exact observation years, smallest practically meaningful effect, historical ATM availability, staffing, exact deadline, computing resources, stakeholders, publication venue, IRB/ethics determination, budget, or formal deliverable acceptance criteria. Those items should be treated as **unspecified**, not inferred. [filecite turn0file1](#)

A realistic baseline execution period is approximately **14–16 weeks** if historical ATM exposure can be resolved quickly. If it cannot, there are only three scientifically honest options: obtain historical records from DFS/platform archives; switch to a prospective outcome window beginning from a known ATM snapshot; or reduce the paper to a more limited design while explicitly weakening exposure claims. The worst option is silently joining a 2026 ATM list to many prior years of crime.

## Research architecture and decision branches

The confirmatory study should formalize the attached plan as a comparison between background crime and incremental site context. [filecite turn0file1](#) Define, for ATM site  $i$  and prediction period  $t$ :

$$Y_{it} = \begin{cases} \text{count of qualifying incidents within } r \text{ of site } i, & \text{count design} \\ 1\{\text{at least one qualifying incident occurs}\}, & \text{binary design} \end{cases}$$

with a preregistered primary radius  $r = 50$  m and sensitivity radii of 25 m and 100 m.

Define lagged background crime as:

$B_{i,t-1}$  = qualifying incidents during the previous period in an outer annulus, e.g. 200–500 m, so that the outcome buffer itself is not recycled into the baseline predictor. This is the central design innovation in the attachments. [Bappee et al. 2019](#)

The confirmatory model ladder should be:

$$M_0 : Y_{it} \sim B_{i,t-1}$$

$$M_1 : Y_{it} \sim B_{i,t-1} + X_i$$

where  $M_1$  is regularized logistic regression for binary outcomes or Negative Binomial regression for counts, and

$$M_2 = f_{\text{boost}}(B_{i,t-1}, X_i)$$

using either XGBoost or LightGBM. Both are established gradient-boosting systems; their use should be described as a flexible tabular benchmark rather than a methodological novelty. [7](#)

For transfer, define:

$$\text{Transfer Gap} = P_{\text{local-full}} - P_{\text{zero-shot}},$$

and

$$\text{Transfer Benefit}(k) = P_{\text{source}+k\% \text{ target}} - P_{k\% \text{ target only}}.$$

The second quantity is essential. Improvement after adding 10% target data is **not evidence that transfer helped** unless source + 10% target also beats a model trained on the same 10% target data alone. This comparison mirrors an important lesson from Bappee et al., where adding suitable target data materially changed cross-domain performance and poorly related sources could produce negative transfer. [8](#)

The overall branching logic should be frozen before the main analysis:



| Situation discovered during execution                                                           | Correct branch                        | Scientific status                   | What not to do                                                   |
|-------------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------------|------------------------------------------------------------------|
| Environmental lift is robustly positive                                                         | Proceed to transfer                   | Core hypothesis supported           | Do not skip M0 or local-full comparator                          |
| Environmental lift is precisely negligible                                                      | Background-dominant result            | Publishable null/equivalence result | Do not hunt features/models until something becomes positive     |
| Environmental lift estimate is near zero but imprecise                                          | Underpowered/inconclusive             | Needs redesign or more data         | Do not label it “no effect”                                      |
| Lift appears in one city only                                                                   | Heterogeneity result                  | Potentially interesting             | Do not claim universal ATM-specific risk                         |
| Zero-shot poor, source + target beats target-only                                               | Useful few-shot transfer              | Strong practical result             | Do not obscure target-only baseline                              |
| Source + target does not beat target-only                                                       | No transfer benefit                   | Negative but informative            | Do not describe simple target calibration as successful transfer |
| Source data makes target prediction worse                                                       | Negative transfer                     | Important result                    | Do not average it away across cities                             |
| Only existing ATM sites are available for decision support                                      | Risk ranking/site evaluation          | Valid extension                     | Do not call it greenfield placement                              |
| Lisbon's 107 observations cannot be verified as exposed attacked and exposed non-attacked sites | Use Lisbon only as motivation/context | Safest                              | Do not reintroduce inferred labels into supervision              |

The last Lisbon branch is particularly important. The detailed attachment identifies the earlier paper's central validity problem as the use of geography-informed inferred labels followed by modeling with overlapping geographical predictors; it recommends retaining Lisbon only as contextual evidence unless a valid observed risk set can be reconstructed. [filecite5turn0file1](#)

## Phase-by-phase execution protocol

The staffing structure is unspecified in the attachments. The instructions below therefore use **functional roles**: PI/lead researcher, data/GIS analyst, statistical/ML analyst, crime-domain reviewer, and QA/reproducibility reviewer. One person can hold several roles in a low-resource project, but the primary crime codebook and a sample of event–ATM matches should ideally receive an independent second review.

## Freeze the research construct

| Item                    | Execution instructions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step-by-step actions    | <b>Step 1:</b> Create <code>study_spec_v1.0</code> before inspecting model performance. <b>Step 2:</b> Freeze the population as bank-owned/in-bank ATM <b>sites</b> , matching the DFS coverage limitation. <b>Step 3:</b> Freeze city boundaries using geographic polygons rather than relying only on address strings. <b>Step 4:</b> Write the exact primary outcome definition and an inclusion/exclusion codebook. <b>Step 5:</b> Freeze the primary 50 m buffer, 25/100 m sensitivities, and background annulus. <b>Step 6:</b> Choose ATM-year or another fixed ATM-period as the unit. <b>Step 7:</b> Freeze M0/M1/M2, holdout hierarchy, primary metric, transfer ladder, and statistical uncertainty procedure. <b>Step 8:</b> Define the smallest effect worth continuing to transfer analysis. <b>Step 9:</b> Timestamp the specification and record all later deviations as exploratory. |
| Plausible cases         | <b>Outcome harmonizes well:</b> proceed unchanged. <b>Larceny/theft descriptions are incompatible:</b> narrow the primary endpoint and move broad theft to sensitivity analysis. <b>One city lacks defensible matching:</b> drop that city rather than invent a translation. <b>Repeated incidents are common:</b> prefer a count/NB formulation. <b>Positive counts are mostly one per site-period and zeros dominate:</b> binary is likely more stable.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Responsible roles       | PI owns construct and claims; crime-domain reviewer approves codebook; methods lead approves unit/model/metric choices.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Required resources      | Attachments; official police dictionaries; one-page protocol; version-controlled repository; decision log.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Estimated time          | 3–7 working days.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Success criteria        | Every primary choice is written down before confirmatory modeling; terminology distinguishes “ATM-site crime” from “ATM-machine attack”; no unresolved city-specific category mapping remains.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Key risk and mitigation | <b>Risk:</b> post-hoc outcome/model shopping. <b>Mitigation:</b> signed/tagged specification and separate confirmatory versus exploratory folders.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

The attached plan already recommends a narrow acquisitive/physical-site outcome, ATM × period unit, 50 m primary radius, 25/100 m sensitivities, background crime baseline, 10–20 common covariates, AUPRC for binary prediction, logistic/NB and boosting, and leave-one-city-out external testing. Those should be treated as the starting defaults unless the data audit demonstrates that one is infeasible. `filecite@turn0file1`

## Build the ATM-site and outcome panel

| Item                    | Execution instructions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step-by-step actions    | <p><b>Step 1:</b> Download and checksum the DFS ATM data. <b>Step 2:</b> preserve raw institution name, address, city, coordinates, and all source fields. <b>Step 3:</b> Normalize addresses without deleting original values. <b>Step 4:</b> Assign municipal boundaries spatially. <b>Step 5:</b> create a stable <code>site_id</code>; where multiple records occupy essentially the same physical facility, create a <code>site_cluster_id</code> as well. <b>Step 6:</b> investigate historical monthly versions. Check the dataset changelog/archive if exposed, then contact NY DFS/data portal administrators for prior extracts if necessary. <b>Step 7:</b> derive <code>active_from</code>, <code>active_to</code>, and active days/months where evidence exists. <b>Step 8:</b> ingest each city's crime data and map native offense fields into the frozen codebook. <b>Step 9:</b> project coordinates into appropriate metric coordinate systems and calculate event-to-site distances. <b>Step 10:</b> aggregate target events to ATM-period. <b>Step 11:</b> calculate lagged background incidents in the outer annulus separately. <b>Step 12:</b> freeze an audit sample and manually inspect it.</p> |
| Exposure branches       | <p><b>Historical ATM snapshots complete:</b> build dynamic ATM-months, aggregate to ATM-years, and use exposure time as an offset for counts when necessary. <b>Partial:</b> use only the common period supported by snapshots. <b>No history:</b> either create a prospective cohort from the known current snapshot or use a clearly labeled weaker recent-retrospective analysis; do not imply known historical exposure.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Matching branches       | <p><b>One event, one nearby site:</b> normal assignment. <b>Several nearby ATMs:</b> retain match multiplicity; primary analysis may use "any site within radius," but run nearest-site-only sensitivity. <b>Co-located ATM records:</b> collapse/cluster at the physical-site level to avoid artificial duplicate sites. <b>Coordinates missing/improbable:</b> geocode only using a documented source, assign a quality flag, and exclude unresolved records from the confirmatory sample.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| QA branches             | <p>If manual inspection reveals substantial transactional fraud, customer-account compromise, or unrelated incidents, tighten the codebook. If errors concentrate near the 50 m boundary, emphasize 25/50/100 m sensitivity. If a city's offense descriptions are too coarse, narrow the common outcome or drop that city.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Responsible roles       | <p>Data/GIS analyst builds panel; crime reviewer validates offense mapping; QA reviewer audits records; PI approves exposure assumptions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Required resources      | <p>Official ATM/crime extracts; GIS tooling; data dictionary; source manifest; manual-audit form.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Estimated time          | <p>2–4 weeks, potentially longer if historical ATM records require agency correspondence.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Success criteria        | <p>Every modeled ATM-period has documented exposure status; target and background crime are distinct; event matching is reproducible; audit errors are understood and corrected before modeling.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Key risk and mitigation | <p><b>Risk:</b> exposure misclassification. <b>Mitigation:</b> historical snapshots/common-window restriction/prospective cohort, plus a sensitivity analysis that shows how results change under alternate exposure assumptions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

The DFS source is particularly suitable for a common denominator because it identifies statewide individual bank-owned/in-bank ATM locations and is posted monthly, but its current-period metadata and omission of independent ATMs must become explicit study limitations. <sup>9</sup> Socrata now supports dataset-version archives where a publisher has enabled archiving, but whether this particular New York dataset has usable past archives must be verified rather than assumed. <sup>10</sup>

For NYC, the historic complaint table is complaint-level and currently contains more than ten million rows; its live preview includes complaints dated through December 31, 2025. The landing-page prose still contains an outdated reference to 2019, illustrating why the pipeline should calculate actual minimum/maximum event dates instead of trusting descriptive text. <sup>11</sup> The 2026 year-to-date complaint dataset was updated July 27, 2026. <sup>12</sup>

For Rochester, another metadata inconsistency needs to be explicitly checked: the public crime-mapping page says Part I mapping data are available since 2011, while the underlying ArcGIS service description says “2000-Present”; importantly, the actual historical layer is labeled “2011 to Present.” Use the queried layer dates, not the broad service description, and record the discrepancy. <sup>13</sup>

### Create the common city-independent feature layer

| Item                     | Execution instructions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step-by-step actions     | <b>Step 1:</b> create a common feature dictionary before extraction. <b>Step 2:</b> retain only information that would have been available before the prediction period. <b>Step 3:</b> compute road accessibility and street-network measures using one definition in all cities. <b>Step 4:</b> attach EPA Smart Location variables by block group where appropriate. <b>Step 5:</b> attach a fixed, ex-ante ACS vintage for socioeconomic controls. <b>Step 6:</b> compute ATM density and nearest-ATM distance from exposure-valid ATM records. <b>Step 7:</b> calculate lagged crime only from prior periods. <b>Step 8:</b> perform missingness, range, distribution, and redundancy checks. <b>Step 9:</b> lock the primary feature matrix before outcome-model tuning. |
| Primary feature families | Ambient background crime; distance/access to major roads; intersection/street density; commercial/activity intensity; population/employment density; destination accessibility; transit accessibility; ATM density/competition; guardianship proxy such as police-station distance; limited socioeconomic controls.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Cases                    | <b>Variable present consistently:</b> keep. <b>Equivalent concept but inconsistent definition:</b> replace with a common source/definition. <b>Only one city has it:</b> exclude from primary transfer model and optionally analyze separately. <b>Feature measured after prediction period:</b> exclude as leakage. <b>High missingness:</b> repair from the same authoritative source or drop rather than city-specific imputation. <b>Highly redundant features:</b> keep the more interpretable/common measure.                                                                                                                                                                                                                                                            |
| Responsible roles        | GIS/data analyst extracts; methods analyst checks distributions; PI approves final schema.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Required resources       | EPA SLD, Census ACS/TIGER, ATM panel, road/network layer, spatial processing code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| Item                    | Execution instructions                                                                                                                                                                                 |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Estimated time          | 2–3 weeks, overlapping with the end of panel construction.                                                                                                                                             |
| Success criteria        | Exactly the same feature definitions and transformations are executable in every city; all predictors have provenance and temporal cutoffs; no target-period crime leaks into covariates.              |
| Key risk and mitigation | <b>Risk:</b> apparently good transfer created by target/future information. <b>Mitigation:</b> source manifest includes “data vintage,” “available-at date,” and prediction cutoff for every variable. |

EPA's Smart Location Database is well suited to this purpose because it provides more than 90 built-environment/location-efficiency attributes at census-block-group scale across most of the United States; however, the current version is the 2021 release, so it should be treated as a fixed contextual layer rather than a contemporaneous 2026 measure. <sup>14</sup> ACS five-year detailed estimates are available down to block-group geography, and TIGER/Line provides official geographic boundaries and road layers that can be linked to Census data. <sup>15</sup>

#### Establish whether an incremental research result exists

| Item                 | Execution instructions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step-by-step actions | <b>Step 1:</b> create temporal and spatial splits before training. <b>Step 2:</b> fit M0 with lagged background crime only. <b>Step 3:</b> fit M1 with the common environmental feature set. <b>Step 4:</b> fit M2 with the same inputs using boosting. <b>Step 5:</b> keep all imputation, scaling, feature selection, weighting, tuning, and calibration inside training folds. <b>Step 6:</b> score only untouched observations. <b>Step 7:</b> calculate paired uncertainty for M1–M0 and M2–M0. <b>Step 8:</b> examine calibration and top-decile capture, not just discrimination. <b>Step 9:</b> make the go/no-go decision from the preregistered primary endpoint. |
| Outcome branches     | <b>Clear positive incremental value:</b> proceed to transfer. <b>Estimate centered near zero with narrow CI inside a prespecified “negligible” margin:</b> support the background-dominant/null conclusion. <b>Near-zero estimate with wide CI:</b> call it inconclusive, not null. <b>Strong value only in one city:</b> treat heterogeneity as the result. <b>Only boosting improves:</b> retain nonlinear model but make “nonlinear incremental signal” the interpretation rather than claiming all models support it. <b>Poor calibration despite useful ranking:</b> report ranking performance but do not present raw probabilities as trustworthy absolute risks.    |
| Responsible roles    | Statistical/ML analyst leads; PI enforces confirmatory gate; reproducibility reviewer checks split isolation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Required resources   | Logistic/NB implementation; XGBoost or LightGBM; calibration tools; blocked resampling; experiment registry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Estimated time       | 2–3 weeks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Item                    | Execution instructions                                                                                                                                  |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Success criteria        | Held-out environmental lift is quantified with uncertainty and compared to M0; no metric is selected after viewing results.                             |
| Key risk and mitigation | <b>Risk:</b> falsely optimistic random cross-validation. <b>Mitigation:</b> use future-time and spatial/city blocking aligned with intended deployment. |

This split strategy is not cosmetic. For spatially or temporally structured data, ordinary random cross-validation can underestimate prediction error; blocked validation is specifically recommended when the intended prediction is to new locations, times, or predictor regimes. <sup>16</sup>

### Perform city-transfer experiments

For each target  $T$ , define a **final target test period first**. Never draw the “10% target” adaptation sample from that test set. The 10% and 25% target labels must come from earlier target periods that would genuinely have been available before the evaluation period.

| Experiment          | Training data                                      | Test data                     | Purpose                                   |
|---------------------|----------------------------------------------------|-------------------------------|-------------------------------------------|
| Local-full          | All eligible prior target data                     | untouched later target period | Upper local benchmark                     |
| Zero-shot           | One source city                                    | same target test              | Direct transportability                   |
| Multi-source        | Other two cities                                   | same target test              | Whether pooling sources improves transfer |
| Source + 10% target | Source city/cities + sampled earlier target labels | same target test              | Low-label adaptation                      |
| 10% target only     | Exactly the same target subset                     | same target test              | Transfer-benefit control                  |
| Source + 25% target | Source city/cities + sampled earlier target labels | same target test              | Larger adaptation                         |
| 25% target only     | Exactly the same target subset                     | same target test              | Larger target-only control                |

For the 10% and 25% experiments, repeat target-subset sampling across multiple fixed seeds and report the distribution or confidence interval, not a lucky single draw.

| Observed case                                                    | Required interpretation/action                                |
|------------------------------------------------------------------|---------------------------------------------------------------|
| Zero-shot nearly matches local-full                              | Strong transportability; quantify small transfer gap          |
| Zero-shot weak, but source + target materially beats target-only | Useful transfer; estimate how much local labeling is required |

| Observed case                                                  | Required interpretation/action                                                            |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Source + target improves over zero-shot but equals target-only | Local data help, but source knowledge adds no demonstrated value                          |
| Target-only beats source + target                              | Negative transfer; diagnose source-target mismatch                                        |
| One source helps and another hurts                             | Do not pool blindly; report source-specific compatibility                                 |
| Both sources together worse than best individual source        | Multi-source negative transfer; retain best-source result and explain heterogeneity       |
| Feature ranges barely overlap between cities                   | Mark extrapolation explicitly; do not treat zero-shot estimates as ordinary interpolation |
| Transfer differs by outcome/radius                             | Keep the preregistered primary result; use alternate results only as sensitivity analyses |

Bappee et al. provide a particularly relevant precedent: their source-only Toronto and Vancouver models generally lost performance on Halifax, target-inclusive configurations performed better, and adding a distant source did not necessarily improve a multi-source model. <sup>8</sup> This is exactly why both target-only controls and negative-transfer reporting belong in the paper.

### Interpret only externally validated relationships

| Item                 | Execution instructions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step-by-step actions | <b>Step 1:</b> interpret only models that passed the prediction gate. <b>Step 2:</b> report logistic/NB coefficients and uncertainty where applicable. <b>Step 3:</b> for boosting, compute SHAP or another prespecified attribution method. <b>Step 4:</b> pair importance with effect-shape plots such as accumulated local effects/partial dependence where appropriate. <b>Step 5:</b> repeat explanation calculations by city. <b>Step 6:</b> compare sign, approximate nonlinear shape, ranking, and bootstrap stability. <b>Step 7:</b> label each variable stable, partially stable, reversed, or unsupported. <b>Step 8:</b> use causal language only if the design truly supports it—which this observational prediction design generally will not. |
| Cases                | <b>Same sign/shape across cities:</b> candidate transportable risk marker. <b>Same sign but different magnitude:</b> partially transportable. <b>Sign reversal:</b> context-specific relationship or proxy/confounding candidate. <b>High SHAP importance but unstable across resamples:</b> do not elevate it as a substantive finding. <b>Strong local effect but no external performance:</b> explanation is local, not transportable.                                                                                                                                                                                                                                                                                                                     |
| Responsible roles    | Methods analyst generates explanations; crime-domain reviewer assesses substantive plausibility; PI controls claims.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Required resources   | Coefficient tables, SHAP implementation, effect plots, bootstrap results.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Estimated time       | 1–2 weeks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Item                    | Execution instructions                                                                                                                                                                                     |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Success criteria        | Every substantive feature claim is connected to a model with demonstrated held-out/external performance and a reproducibility/stability check.                                                             |
| Key risk and mitigation | <b>Risk:</b> interpreting model artifacts as criminological mechanisms. <b>Mitigation:</b> use language such as “predictive association,” compare cities, and triangulate statistical and flexible models. |

SHAP is an appropriate tool for attribution, but its original formulation explains model predictions rather than establishing causality. <sup>17</sup> Recent crime-prediction research likewise shows that explanation methods can differ locally even when global rankings are broadly similar, reinforcing the need for stability checks instead of treating one SHAP ranking as ground truth. <sup>18</sup>

### Decide whether placement belongs in the paper

The placement analysis should remain gated behind successful risk validation, exactly as the attachments recommend. [filecite turn0file1](#)

| Data situation                                                                            | Action                                                                                                                    |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Genuine candidate locations where no ATM currently exists + demand + accessibility + cost | Add an optional multi-objective/Pareto analysis                                                                           |
| Candidate locations exist but no demand measure                                           | Do not optimize placement; at most show risk screening                                                                    |
| Only existing ATM sites are available                                                     | Call the extension “ATM-site evaluation” or “security-aware site assessment”                                              |
| Risk model has not survived external validation                                           | Do not add placement at all                                                                                               |
| Stakeholders demand one composite score                                                   | Present transparent weights and sensitivity as a secondary decision illustration, not the paper's scientific contribution |

A simple formulation is:

$$\min_x R(x)$$

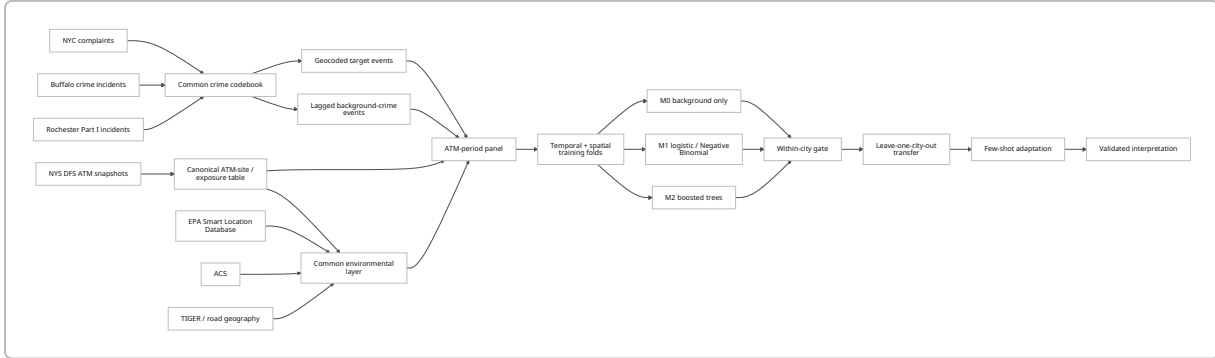
subject to

$$A(x) \geq A_{\min}, \quad D(x) \geq D_{\min},$$

or, if risk, access, and cost are genuinely measured, retain nondominated sites on a Pareto frontier. This is methodologically cleaner than resurrecting a fuzzy/Choquet aggregation solely to create apparent novelty. [filecite turn0file1](#)

## Data collection, analysis, and validation specification

The raw-data pipeline should be reproducible from immutable extracts:



### Prioritized data and evidence sources

| Priority         | Source                                      | Recommended use                                           | Important constraint                                                                                                                                                |
|------------------|---------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary official | New York State DFS Bank-Owned ATM Locations | ATM denominator and coordinates                           | Current-period dataset; bank-owned/in-bank only; monthly posting. <sup>9</sup>                                                                                      |
| Primary official | NYPD Historic + Current Complaint Data      | NYC incident records and detailed offense coding          | Complaint-level rather than dedicated ATM-attack records; use occurrence/offense fields carefully and follow dataset footnotes. <sup>5</sup>                        |
| Primary official | Buffalo Crime Incidents                     | Buffalo point/event data                                  | Verify exact minimum/maximum dates, coordinates, missingness and category stability programmatically; portal indicates daily-updated Crime Incidents. <sup>19</sup> |
| Primary official | Rochester RPD Part I Crime service          | Rochester robbery/burglary/larceny etc.                   | Historical layer is labeled 2011-present; reconcile service-level “2000-present” metadata before analysis. <sup>20</sup>                                            |
| Primary official | EPA Smart Location Database v3              | Common built environment, density, accessibility, transit | Current SLD version dates to 2021; treat as fixed-context data. <sup>14</sup>                                                                                       |
| Primary official | Census ACS five-year                        | Block-group socioeconomic controls                        | Detailed estimates available at block-group scale; select an ex-ante vintage. <sup>21</sup>                                                                         |
| Primary official | Census TIGER/Line                           | Boundaries, roads, geographic keys                        | Geographic rather than demographic data; link using Census codes. <sup>22</sup>                                                                                     |

| Priority          | Source                             | Recommended use                                        | Important constraint                                                                               |
|-------------------|------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Original research | Bappee et al.                      | Transfer-experiment precedent and target-data controls | Shows source-target distribution differences and negative-transfer possibility. <sup>23</sup>      |
| Original research | Zhou et al.                        | Advanced-domain-adaptation comparison/future work      | Useful related work; unnecessary for initial implementation. <sup>2</sup>                          |
| Original research | Barnum et al.                      | Cross-jurisdiction environmental-risk rationale        | Similar risk factors can operate differently across urban environments. <sup>24</sup>              |
| Original research | Kronkvist et al.                   | Multi-city common-feature and prior-crime comparisons  | Demonstrates common OSM/register place features and models with/without prior crime. <sup>25</sup> |
| Original methods  | Roberts et al.; Pohjankukka et al. | Spatial/temporal validation                            | Supports blocked rather than ordinary random CV for dependent geographic data. <sup>16</sup>       |
| Original methods  | Saito & Rehmsmeier                 | Binary imbalance evaluation                            | Supports PR/AUPRC emphasis when positives are rare. <sup>26</sup>                                  |
| Original methods  | Chen & Guestrin; Ke et al.         | Flexible tabular models                                | Original XGBoost and LightGBM references. <sup>7</sup>                                             |
| Original methods  | Lundberg & Lee                     | Model attribution                                      | Original SHAP framework. <sup>17</sup>                                                             |

**Recommended target-codebook rule.** The main outcome should include only events that can defensibly be interpreted as physical/acquisitive incidents in the ATM micro-environment. Purely transactional fraud, bank-account compromise, forgery, identity theft, and electronically mediated loss should not be mixed into the physical-site endpoint merely to enlarge the sample. The attached plan reaches the same conclusion after noting the heterogeneity of the existing NYC ATM-premise complaints. [filecite turn0file1](#)

The event audit should be expanded beyond the original suggestion of approximately 100 NYC matches. A stronger implementation is a **stratified review across all included cities**, oversampling ambiguous offense categories and cases near 25/50/100 m thresholds. Record event ID, site ID, distance, native description, common category, fraud/transaction flag, duplicate flag, location plausibility, and reviewer decision. A second reviewer should independently code a subset and disagreements should be adjudicated before the final codebook is locked.

### Recommended panel schema

| Variable group | Example fields                                                                              |
|----------------|---------------------------------------------------------------------------------------------|
| Site identity  | <code>site_id</code> , <code>site_cluster_id</code> , institution, normalized address, city |

| Variable group  | Example fields                                                                                              |
|-----------------|-------------------------------------------------------------------------------------------------------------|
| Exposure        | <code>active_from</code> , <code>active_to</code> , <code>active_days</code> , exposure-source/quality flag |
| Time            | period start/end, prediction cutoff                                                                         |
| Outcome         | target count, target binary, offense-specific counts                                                        |
| Background      | prior-period qualifying crimes in 200–500 m annulus                                                         |
| Roads           | distance to major road, road/intersection density                                                           |
| Activity/access | commercial intensity, destination accessibility, transit accessibility                                      |
| Urban form      | population/employment density, design measures                                                              |
| Guardianship    | police-station/access proxy if consistently available                                                       |
| ATM environment | ATM density, nearest-ATM distance                                                                           |
| Context         | selected income/unemployment or other preregistered nonprotected controls                                   |
| QA              | coordinate quality, offense-codebook version, matching multiplicity, source-version hash                    |

**Validation hierarchy.** The principal result should come from a deployment-realistic split, not random cross-validation. Use training on earlier periods and testing on later periods; add spatial holdouts such as neighborhood/grid/precinct groups; then rotate a completely held-out city. Spatially structured random splits can give optimistic error estimates because neighboring observations are dependent. <sup>16</sup>

All learned preprocessing belongs inside the training folds: missing-value imputation, scaling, feature selection, hyperparameter optimization, class weighting/resampling, and probability calibration. A final test city/time block must never influence these operations.

For binary outcomes, designate **AUPRC as the primary discrimination metric** if the positive class is rare; precision-recall analysis is more informative than relying only on ROC curves under strong class imbalance.

<sup>26</sup> Also report ROC-AUC, Brier score, calibration intercept/slope or calibration plot, top-5% and top-10% recall, precision in the top risk fraction, and lift. For counts, report Negative Binomial/Poisson deviance as appropriate, MAE, calibration of predicted counts, and site-ranking performance.

Do not bootstrap individual ATM-period rows naively when the same ATM appears repeatedly. For uncertainty, resample at least at the **ATM/site level**; where residual spatial dependence remains material, use a spatial-block bootstrap. Model-to-model comparisons should be paired so both models are evaluated on identical held-out observations.

For logistic regression, standardization parameters must be learned from training data only. For zero-shot transfer, the safest primary representation uses physically interpretable common units. A sensitivity analysis may use city-standardized variables, but because target-city feature distributions are then used in constructing the representation, that analysis should be labeled explicitly as a transductive/domain-normalized variant.

The paper should report **distribution shift before transfer**. For every continuous feature, compare ranges, quantiles, missingness, and standardized differences between source and target. A multivariate distance diagnostic can be supplementary. Bappee et al. explicitly examined source-target feature-distribution differences before interpreting transfer performance, and their results show why this matters. <sup>27</sup>

## Risk register, branching matrix, and budget

### Risk register

| Risk                                                     | Likelihood  | Impact   | Trigger/<br>evidence                                                 | Mitigation                                                                                                                          | Owner          |
|----------------------------------------------------------|-------------|----------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Historical ATM exposure cannot be reconstructed          | High        | Critical | Only current DFS list accessible                                     | Search official archive/changelog; contact DFS; restrict common period; prospectively freeze cohort; downgrade retrospective claims | PI + data lead |
| Current ATM records are mistakenly treated as historical | Preventable | Critical | Same ATM universe appears in every historical year                   | Automated exposure assertion; prohibit panel construction until <code>active_from/active_to</code> rule is signed off               | Data lead      |
| Crime construct differs across cities                    | Medium-High | Critical | Large numbers of native categories have no defensible common mapping | Narrow primary endpoint; separate secondary outcomes; drop incompatible city                                                        | Crime reviewer |
| "Near an ATM" is interpreted as "attack on ATM"          | Medium      | High     | Manuscript wording exceeds available police fields                   | Use "ATM-site/micro-environment acquisitive crime" terminology                                                                      | PI             |
| Rare target events produce weak power                    | High        | High     | Extreme zero proportion; unstable coefficients/CIs                   | Consider two-year windows, binary endpoint, reduced feature count, pooled source training; report inconclusive rather than null     | Methods lead   |

| Risk                                                 | Likelihood          | Impact      | Trigger/<br>evidence                                            | Mitigation                                                                                    | Owner        |
|------------------------------------------------------|---------------------|-------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------|--------------|
| Spatial/temporal leakage inflates accuracy           | Medium              | Critical    | Random CV much better than blocked/future holdout               | Make temporal/spatial/city holdouts primary; pipeline all preprocessing inside folds          | Methods lead |
| Outcome accidentally enters background predictor     | Medium              | Critical    | Same event falls in target and lagged/background window         | Enforce time lag and inner exclusion radius programmatically                                  | Data lead    |
| Overlapping ATM buffers share events                 | High in dense areas | Medium      | Event linked to several sites                                   | Record multiplicity; cluster co-located ATMs; nearest-only sensitivity; clustered uncertainty | GIS lead     |
| Source-target feature shift causes negative transfer | Medium-High         | High        | Large distribution mismatch; source+target < target-only        | Report negative transfer; compare individual sources; only then consider weighting/adaptation | Methods lead |
| Class imbalance makes accuracy misleading            | High                | High        | Majority-class accuracy dominates                               | Use AUPRC and ranking/calibration metrics rather than accuracy as primary                     | Methods lead |
| Model explanation is unstable                        | Medium              | Medium-High | SHAP/effect ranks change across resamples/cities                | Bootstrap explanation stability; report only externally validated relationships               | Methods lead |
| Official source metadata contain inconsistencies     | Medium              | Medium      | Rochester/NYC descriptive dates conflict with actual layer/rows | Calculate date ranges and schema directly; version raw files; document anomalies              | Data lead    |
| Public datasets update during the project            | High                | Medium      | Row counts or records change                                    | Archive immutable raw snapshots with checksums and retrieval timestamp                        | Repro lead   |

| Risk                                                                  | Likelihood | Impact | Trigger/<br>evidence                                      | Mitigation                                                                                                           | Owner              |
|-----------------------------------------------------------------------|------------|--------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------|
| Scope creep into deep learning or placement                           | Medium     | High   | Core validation incomplete while new components are added | Enforce go/no-go gates; move additions to future work                                                                | PI                 |
| Protected/social attributes produce problematic operational targeting | Medium     | High   | Risk score driven by sensitive demographics               | Exclude protected attributes from operational model unless specifically justified; discuss fairness and intended use | PI/ethics reviewer |
| One city fails late                                                   | Medium     | High   | Data-quality gate fails after substantial work            | Build two-city and NYC-only fallbacks from the start                                                                 | PI                 |

The two source-quality issues already visible in official portals—Rochester’s differing date descriptions and NYC’s stale prose alongside current records—make the raw-source manifest and automated date assertions essential rather than optional. <sup>28</sup>

### Branching decision matrix

| Decision                           | Preferred choice                                                                   | Switch when                               | Fallback                             |
|------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------|
| Count vs binary outcome            | Negative Binomial count if repeated site-period incidents contain real information | Counts are overwhelmingly 0/1 or unstable | Binary logistic + boosted classifier |
| Three vs two cities                | NYC + Buffalo + Rochester                                                          | One city fails outcome/exposure gate      | Best two compatible cities           |
| Historical vs prospective exposure | Historical dynamic panel                                                           | Reliable past ATM snapshots unavailable   | Prospective cohort                   |
| 50 m outcome radius                | Keep as confirmatory                                                               | Never switch because results look better  | 25/100 m only as sensitivities       |
| Feature count                      | 10–20 common variables                                                             | Events-per-parameter/power insufficient   | Smaller theory-driven set            |
| Imbalance method                   | Class weights/proper loss                                                          | Only reconsider if diagnostics justify    | Training-fold-only sampling          |

| Decision        | Preferred choice                                    | Switch when                               | Fallback                                      |
|-----------------|-----------------------------------------------------|-------------------------------------------|-----------------------------------------------|
| Flexible model  | One of XGBoost or LightGBM                          | Implementation/<br>reproducibility issue  | The other boosting<br>implementation          |
| Transfer method | Direct source/pooled/few-<br>shot ladder            | Only after baseline<br>ladder complete    | Optional TrAdaBoost/<br>weighting exploratory |
| Explainability  | Coefficients + SHAP/effect<br>curves                | Prediction fails<br>external validation   | Do not interpret ML<br>feature importance     |
| Placement       | Pareto/multi-objective only<br>with true candidates | Candidate/demand/<br>cost data incomplete | Stop at site-risk<br>evaluation               |

**Budget.** No budget is supplied in either attachment, so the following figures are planning scenarios rather than documented project commitments. Core source data are publicly accessible through the cited government portals; the major cost is labor rather than data licensing. <sup>29</sup>

| Phase                           | Low scenario*      | Medium scenario**        | High scenario***         |
|---------------------------------|--------------------|--------------------------|--------------------------|
| Protocol/codebook               | \$0–\$300          | \$500–\$1,500            | \$2,000–\$4,000          |
| ATM exposure/panel engineering  | \$0–\$700          | \$2,500–\$6,000          | \$7,000–\$14,000         |
| Crime harmonization + manual QA | \$0–\$500          | \$2,000–\$5,000          | \$6,000–\$12,000         |
| Common GIS/features             | \$0–\$500          | \$1,500–\$4,000          | \$4,000–\$9,000          |
| Within-city models/validation   | \$0–\$500          | \$2,000–\$5,000          | \$5,000–\$10,000         |
| Transfer experiments            | \$0–\$400          | \$1,500–\$4,000          | \$4,000–\$8,000          |
| Interpretation/sensitivity      | \$0–\$300          | \$1,000–\$3,000          | \$3,000–\$7,000          |
| Reproducibility + manuscript    | \$0–\$500          | \$2,000–\$5,000          | \$5,000–\$12,000         |
| <b>Approximate total</b>        | <b>\$0–\$3,700</b> | <b>\$13,000–\$33,500</b> | <b>\$36,000–\$76,000</b> |

\* Low: researcher labor is already funded/not monetized; open-source software and local compute.

\* Medium: substantial part-time RA/GIS/methods support and modest compute/storage.

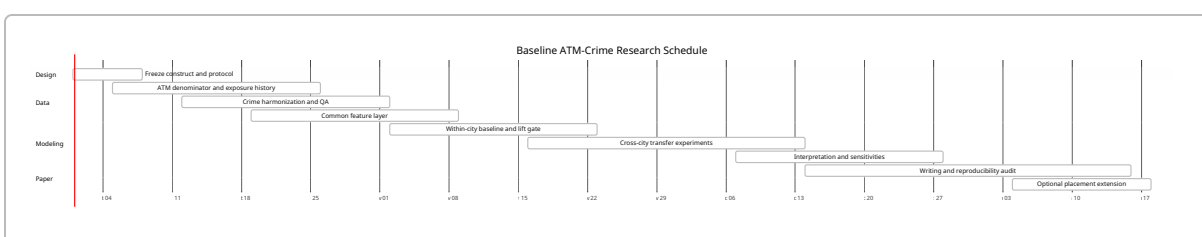
\* High: dedicated analyst/RA time, independent coding/QA, statistical consultation, and stronger reproducibility support. Institutional overhead, publication fees, conference travel, and salaries already committed to staff are not specified\* and are excluded.

The low scenario is viable technically because this is primarily a tabular/GIS project rather than a GPU-intensive deep-learning project. The high scenario mainly buys labor, independent validation, and faster execution—not a fundamentally different scientific design.

## Timeline and consolidated checklist

The following baseline schedule begins **October 1, 2026** and assumes historical ATM exposure is resolved by approximately the end of the third week. The exact project deadline is unspecified in the attachments. If historical exposure cannot be recovered, a strong prospective validation will necessarily extend beyond this schedule.

| Calendar period | Main activity                                               | Gate/output                  |
|-----------------|-------------------------------------------------------------|------------------------------|
| Oct 1–7         | Freeze protocol, crime construct, codebook draft            | <code>study_spec_v1.0</code> |
| Oct 5–26        | ATM data, deduplication, exposure-history investigation     | Exposure decision            |
| Oct 12–Nov 2    | Crime ingestion, crosswalk, event/site QA                   | Validated common outcome     |
| Oct 19–Nov 9    | EPA/ACS/TIGER/common GIS feature layer                      | Frozen feature matrix        |
| Nov 2–23        | M0/M1/M2, temporal/spatial validation                       | Incremental-signal gate      |
| Nov 16–Dec 14   | Leave-one-city-out + multi-source + 10%/25% transfer        | Transfer table               |
| Dec 7–28        | Stability, SHAP/effect curves, radius/outcome sensitivities | Interpretation package       |
| Dec 14–Jan 15   | Manuscript, appendices, reproducibility audit               | Submission-ready core paper  |
| Jan 4 onward    | Optional placement layer only if all core gates passed      | Optional extension           |



### Consolidated execution checklist

| Check                    | Required before proceeding                                                                |
|--------------------------|-------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | Population explicitly says bank-owned/in-bank ATM <b>sites</b> , not all ATMs in New York |
| <input type="checkbox"/> | Primary outcome has written inclusions/exclusions                                         |
| <input type="checkbox"/> | Transactional fraud/forgery/account compromise handled separately                         |

| Check                    | Required before proceeding                                                                       |
|--------------------------|--------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | Primary unit, prediction period, radius, and outer background annulus frozen                     |
| <input type="checkbox"/> | Exact city boundary rule frozen                                                                  |
| <input type="checkbox"/> | DFS raw extract archived with retrieval date and checksum                                        |
| <input type="checkbox"/> | Historical ATM availability investigated and documented                                          |
| <input type="checkbox"/> | No ATM-period exists outside defensible site exposure                                            |
| <input type="checkbox"/> | Crime datasets archived separately by city and version                                           |
| <input type="checkbox"/> | Native crime codes mapped to one common codebook                                                 |
| <input type="checkbox"/> | Event dates and coordinate ranges programmatically validated                                     |
| <input type="checkbox"/> | Duplicate/co-located ATM rule implemented                                                        |
| <input type="checkbox"/> | Event-to-multiple-ATM linkage multiplicity recorded                                              |
| <input type="checkbox"/> | Stratified event-ATM manual QA completed                                                         |
| <input type="checkbox"/> | Core features use one definition/code path in every city                                         |
| <input type="checkbox"/> | Every predictor has a provenance and “available before prediction” check                         |
| <input type="checkbox"/> | Protected demographic variables excluded from the operational core unless specifically justified |
| <input type="checkbox"/> | M0 background-only baseline fitted before claims about environmental risk                        |
| <input type="checkbox"/> | Temporal and spatial holdouts fixed before tuning                                                |
| <input type="checkbox"/> | Preprocessing occurs entirely inside training folds                                              |
| <input type="checkbox"/> | AUPRC/count metric and go/no-go criterion fixed                                                  |
| <input type="checkbox"/> | Confidence intervals accompany environmental lift                                                |
| <input type="checkbox"/> | Precise null distinguished from underpowered/inconclusive result                                 |
| <input type="checkbox"/> | Final target-city test data never used for target adaptation                                     |
| <input type="checkbox"/> | 10%/25% source+target runs compared with identical target-only subsets                           |
| <input type="checkbox"/> | Transfer gap and transfer benefit reported separately                                            |
| <input type="checkbox"/> | Source-target feature distribution shift documented                                              |
| <input type="checkbox"/> | Explainability run only after external-validation gate                                           |
| <input type="checkbox"/> | 25/50/100 m sensitivity completed without changing the confirmatory radius                       |
| <input type="checkbox"/> | Narrow/broad outcome sensitivity completed                                                       |

| Check                    | Required before proceeding                                                                                    |
|--------------------------|---------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | Exposure-assumption sensitivity completed                                                                     |
| <input type="checkbox"/> | All results traceable to data hash, code commit, model seed, and experiment ID                                |
| <input type="checkbox"/> | Placement claims excluded unless genuine non-ATM candidate sites exist                                        |
| <input type="checkbox"/> | Lisbon inferred labels excluded from supervised New York modeling                                             |
| <input type="checkbox"/> | Final paper limitations explicitly include bank-owned-only coverage and police-report measurement limitations |

The intended reporting order should be equally disciplined: **data validity** → **value above background crime** → **future/spatial validation** → **cross-city transfer** → **few-shot recovery** → **interpretation**. That ordering keeps the paper's scientific logic visible and prevents methods from obscuring the main question.

## Documentation templates and final deliverables

The following templates should be created at the start of the project rather than reconstructed during manuscript writing.

### Preregistered study specification

STUDY ID:

VERSION / DATE:

CONFIRMATORY OR EXPLORATORY:

Research question:

Does ATM-site environmental context predict future acquisitive crime beyond lagged surrounding crime, and does that signal transfer across cities?

Population:

Bank-owned/in-bank ATM sites in \_\_\_\_\_

Cities:

\_\_\_\_\_

Observation/exposure period:

\_\_\_\_\_

Primary outcome:

Exact included categories:

Exact excluded categories:

Primary unit:

ATM site × \_\_\_\_\_ period

Primary site radius:

\_\_\_\_\_ meters

Sensitivity radii:

\_\_\_\_\_ / \_\_\_\_\_ meters

Background-crime definition:

Prior period: \_\_\_\_\_

Inner exclusion: \_\_\_\_\_

Outer radius/annulus: \_\_\_\_\_

Core feature set:

\_\_\_\_\_

Primary statistical model:

\_\_\_\_\_

Primary flexible model:

\_\_\_\_\_

Primary baseline:

\_\_\_\_\_

Primary metric:

\_\_\_\_\_

Secondary metrics:

\_\_\_\_\_

Temporal holdout:

\_\_\_\_\_

Spatial holdout:

\_\_\_\_\_

External city holdout:

\_\_\_\_\_

Target adaptation fractions:

\_\_\_\_\_

Primary go/no-go criterion:

\_\_\_\_\_

Smallest effect considered practically meaningful:

\_\_\_\_\_

Sensitivity analyses:

---

Known deviations from attached redesign plan:

---

### Raw-data/source manifest

| Field                        | Required entry                                  |
|------------------------------|-------------------------------------------------|
| Source ID                    | Short immutable project name                    |
| Provider                     | DFS / NYPD / Buffalo / Rochester / EPA / Census |
| Official dataset name        | Exact published title                           |
| Dataset identifier           | Socrata/ArcGIS/API identifier where relevant    |
| Retrieved                    | Timestamp and time zone                         |
| File hash                    | SHA-256                                         |
| Row count                    | Before filters                                  |
| Minimum event/exposure date  | Calculated from data                            |
| Maximum event/exposure date  | Calculated from data                            |
| Coordinate CRS               | Source and analysis CRS                         |
| Filters                      | Exact query/filter                              |
| Missing coordinates          | Count/%                                         |
| Data vintage                 | Publication/reference period                    |
| Available-before-prediction? | Yes/no + reason                                 |
| Raw file modified?           | Must be "No"; transformations create new files  |
| Notes                        | Metadata conflicts, exclusions, quality issues  |

This manifest is particularly important because the official sources themselves can update and, as noted above, some descriptive metadata do not perfectly match the underlying records. <sup>30</sup>

### Crime codebook

| City      | Native offense code | Native description | Common category                     | Primary included? | Financial-only? | Ambiguity rule | Reviewer |
|-----------|---------------------|--------------------|-------------------------------------|-------------------|-----------------|----------------|----------|
| NYC       |                     |                    | robbery/<br>burglary/<br>theft/etc. | Y/N               | Y/N             |                |          |
| Buffalo   |                     |                    |                                     |                   |                 |                |          |
| Rochester |                     |                    |                                     |                   |                 |                |          |

The final appendix should publish this crosswalk so that reviewers can see exactly what “ATM-site acquisitive crime” means in each jurisdiction.

#### Event-to-ATM audit record

```

AUDIT ID:
CITY:
EVENT ID:
ATM SITE ID:
SITE CLUSTER ID:

Event date:
Native offense:
Common offense:
Event coordinates:
ATM coordinates:
Distance in meters:

Within primary radius?      Y / N
Within sensitivity radius?  25 / 50 / 100 / none
Physical/acquisitive event? Y / N / uncertain
Transactional/financial?   Y / N / uncertain
Duplicate complaint/event?  Y / N / uncertain
Multiple ATM matches?      Y / N
Nearest ATM assignment:     _____

Reviewer decision:
KEEP / EXCLUDE / ADJUDICATE

Reason:
_____

```

#### Experiment run card

RUN ID:  
DATE:  
GIT COMMIT:  
DATASET HASH / PANEL VERSION:  
CODEBOOK VERSION:

Target city:  
Source city/cities:

Training dates:  
Adaptation dates:  
Final test dates:

Adaptation fraction:  
Random seed / sampling replicate:

Model:  
Hyperparameters:

Feature set version:  
Imputation fitted on:  
Scaling fitted on:  
Calibration fitted on:

Primary metric:  
AUPRC / Deviance =

Secondary:  
ROC-AUC =  
Brier =  
Top-10% recall =  
Top-10% precision =  
Lift =  
Calibration =

Baseline M0 result:  
Environmental M1/M2 result:  
Delta from M0:  
95% CI:

Transfer gap:  
Transfer benefit:

Pass/fail predefined gate:  
Reason:

Known warnings:

### Core results table

| Target city | Local-full | Zero-shot best source | Other cities pooled | +10% target | 10% target only | +25% target | 25% target only | Transfer gap | Transfer benefit |
|-------------|------------|-----------------------|---------------------|-------------|-----------------|-------------|-----------------|--------------|------------------|
| NYC         |            |                       |                     |             |                 |             |                 |              |                  |
| Buffalo     |            |                       |                     |             |                 |             |                 |              |                  |
| Rochester   |            |                       |                     |             |                 |             |                 |              |                  |

This should be one of the paper's main tables; the attached plan correctly argues that a clear transfer table is scientifically more valuable than adding several additional classifiers. [filecite.bturn0file1](#)

A second central table should show the **incremental environmental signal**:

| City      | M0 background only | M1 background + environment | M2 boosted | $\Delta M1-M0$ with CI | $\Delta M2-M0$ with CI | Conclusion |
|-----------|--------------------|-----------------------------|------------|------------------------|------------------------|------------|
| NYC       |                    |                             |            |                        |                        |            |
| Buffalo   |                    |                             |            |                        |                        |            |
| Rochester |                    |                             |            |                        |                        |            |

A third should summarize transportable interpretation:

| Feature                  | NYC direction/ shape | Buffalo | Rochester | Stability classification | Interpretation |
|--------------------------|----------------------|---------|-----------|--------------------------|----------------|
| Lagged background crime  |                      |         |           |                          |                |
| Major-road accessibility |                      |         |           |                          |                |
| Commercial intensity     |                      |         |           |                          |                |
| Transit accessibility    |                      |         |           |                          |                |
| ATM density              |                      |         |           |                          |                |

| Feature | NYC direction/<br>shape | Buffalo | Rochester | Stability<br>classification | Interpretation |
|---------|-------------------------|---------|-----------|-----------------------------|----------------|
|---------|-------------------------|---------|-----------|-----------------------------|----------------|

Police/guardianship  
proxy

The final reproducible deliverable should contain the frozen protocol, source manifest, raw-data retrieval instructions, crime codebook, ATM exposure construction, derived-data dictionary, QA report, modeling/validation code, experiment registry, main tables/figures, sensitivity appendix, limitation statement, and manuscript. The principal paper should follow the sequence recommended in the attachment: ATM-risk/background-crime problem, data and exposure definition, common feature design, models, temporal/spatial/external validation, environmental lift, transfer, few-shot recovery, interpretation, limitations, and conclusion. [filecite turn0file1](#)

The final scientific claim should remain deliberately narrow: **the study tests whether bank-owned/in-bank ATM sites contain a reproducible, transportable environmental risk signal beyond surrounding crime concentration**. A positive result, a precise null result, and a finding of strong within-city but poor cross-city transportability are all legitimate outcomes. The redesigned paper becomes weak only if the analysis changes definitions, cities, radii, features, or models after observing which choices produce the most attractive answer.

1 Examining the impact of cross-domain learning on crime prediction | Journal of Big Data | Springer Nature Link

[https://link.springer.com/article/10.1186/s40537-021-00489-9?utm\\_source=chatgpt.com](https://link.springer.com/article/10.1186/s40537-021-00489-9?utm_source=chatgpt.com)

2 Unsupervised Domain Adaptation for Crime Risk Prediction Across Cities

[https://doi.org/10.1109%2FTCSS.2022.3207987?utm\\_source=chatgpt.com](https://doi.org/10.1109%2FTCSS.2022.3207987?utm_source=chatgpt.com)

3 24 The Crime Kaleidoscope: A Cross-Jurisdictional Analysis of Place Features and Crime in Three Urban Environments

[https://www.ojp.gov/ncjrs/virtual-library/abstracts/crime-kaleidoscope-cross-jurisdictional-analysis-place-features-and?utm\\_source=chatgpt.com](https://www.ojp.gov/ncjrs/virtual-library/abstracts/crime-kaleidoscope-cross-jurisdictional-analysis-place-features-and?utm_source=chatgpt.com)

4 9 29 About Bank-Owned ATM Locations in New York State Map | State of New York

[https://data.ny.gov/en/Government-Finance/Bank-Owned-ATM-Locations-in-New-York-State-Map/ygnk-5wb9/about?utm\\_source=chatgpt.com](https://data.ny.gov/en/Government-Finance/Bank-Owned-ATM-Locations-in-New-York-State-Map/ygnk-5wb9/about?utm_source=chatgpt.com)

5 11 30 NYPD Complaint Data Historic | NYC Open Data

[https://data.cityofnewyork.us/Public-Safety/NYPD-Complaint-Data-Historic/qgea-i56i?utm\\_source=chatgpt.com](https://data.cityofnewyork.us/Public-Safety/NYPD-Complaint-Data-Historic/qgea-i56i?utm_source=chatgpt.com)

6 19 data.buffalony.gov

[https://data.buffalony.gov/w/d6g9-xbgu/ixf7-aa9j?cur=6omw1tqotmP&utm\\_source=chatgpt.com](https://data.buffalony.gov/w/d6g9-xbgu/ixf7-aa9j?cur=6omw1tqotmP&utm_source=chatgpt.com)

7 XGBoost | Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining

[https://doi.org/10.1145/2939672.2939785?utm\\_source=chatgpt.com](https://doi.org/10.1145/2939672.2939785?utm_source=chatgpt.com)

8 23 27 Examining the impact of cross-domain learning on crime prediction | Journal of Big Data | Springer Nature Link

<https://link.springer.com/article/10.1186/s40537-021-00489-9>

10 Introducing Dataset Archiving – Data & Insights Client Center

[https://support.socrata.com/hc/en-us/articles/9486838238743-Introducing-Dataset-Archiving?utm\\_source=chatgpt.com](https://support.socrata.com/hc/en-us/articles/9486838238743-Introducing-Dataset-Archiving?utm_source=chatgpt.com)

12 NYPD Complaint Data Current (Year To Date) | NYC Open Data

[https://data.cityofnewyork.us/Public-Safety/NYPD-Complaint-Data-Current-Year-To-Date-/5uac-w243?utm\\_source=chatgpt.com](https://data.cityofnewyork.us/Public-Safety/NYPD-Complaint-Data-Current-Year-To-Date-/5uac-w243?utm_source=chatgpt.com)

13 RPD: Rochester crime mapping | City of Rochester, New York

[https://www.cityofrochester.gov/departments/rochester-police-department/rpd-rochester-crime-mapping?utm\\_source=chatgpt.com](https://www.cityofrochester.gov/departments/rochester-police-department/rpd-rochester-crime-mapping?utm_source=chatgpt.com)

14 Smart Location Mapping | US EPA

[https://www.epa.gov/smartgrowth/smart-location-mapping?utm\\_source=chatgpt.com](https://www.epa.gov/smartgrowth/smart-location-mapping?utm_source=chatgpt.com)

15 21 American Community Survey 5-Year Data (2009-2024)

[https://www.census.gov/data/developers/data-sets/acs-5year.html?utm\\_source=chatgpt.com](https://www.census.gov/data/developers/data-sets/acs-5year.html?utm_source=chatgpt.com)

16 Ecography - Wiley Online Library

[https://nsojournals.onlinelibrary.wiley.com/doi/full/10.1111/ecog.02881?utm\\_source=chatgpt.com](https://nsojournals.onlinelibrary.wiley.com/doi/full/10.1111/ecog.02881?utm_source=chatgpt.com)

17 A Unified Approach to Interpreting Model Predictions

[https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html?utm\\_source=chatgpt.com](https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html?utm_source=chatgpt.com)

18 Comparing XAI techniques for interpreting short-term burglary predictions at micro-places | Computational Urban Science | Springer Nature Link

[https://link.springer.com/article/10.1007/s43762-025-00185-x?utm\\_source=chatgpt.com](https://link.springer.com/article/10.1007/s43762-025-00185-x?utm_source=chatgpt.com)

20 28 RPD/RPD\_Part\_I\_Crime (MapServer)

[https://gis.cityofrochester.gov/arcgis/rest/services/RPD/RPD\\_Part\\_I\\_Crime/MapServer](https://gis.cityofrochester.gov/arcgis/rest/services/RPD/RPD_Part_I_Crime/MapServer)

22 TIGER/Line Geodatabase Documentation

[https://www.census.gov/programs-surveys/geography/technical-documentation/complete-technical-documentation/tiger-geodatabase-file.2025.html?utm\\_source=chatgpt.com](https://www.census.gov/programs-surveys/geography/technical-documentation/complete-technical-documentation/tiger-geodatabase-file.2025.html?utm_source=chatgpt.com)

25 Predicting Public Violent Crime Using Register and OpenStreetMap Data: A Risk Terrain Modeling Approach Across Three Cities of Varying Size | Applied Spatial Analysis and Policy | Springer Nature Link

[https://link.springer.com/article/10.1007/s12061-024-09609-3?utm\\_source=chatgpt.com](https://link.springer.com/article/10.1007/s12061-024-09609-3?utm_source=chatgpt.com)

26 The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets | PLOS One

[https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0118432&utm\\_source=chatgpt.com](https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0118432&utm_source=chatgpt.com)